AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 20 July 2026 01:19 AM

Developer: Pritiranjan Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report evaluates organizational workforce data across 20 records to assess employee performance, compensation structures, and data integrity. The workforce demonstrates an exceptionally strong alignment between tenure, compensation, and performance. However, minor data quality anomalies require immediate remediation to ensure accurate future planning.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjan Rout
Version	2.0.0
Generated On	20 July 2026 01:19 AM

DATASET SUMMARY

Metric	Value

Rows	20
Columns	10
Health Score	95%
Duplicate Rows	0

DATASET HEALTH SCORE

Overall Health Score : 95%

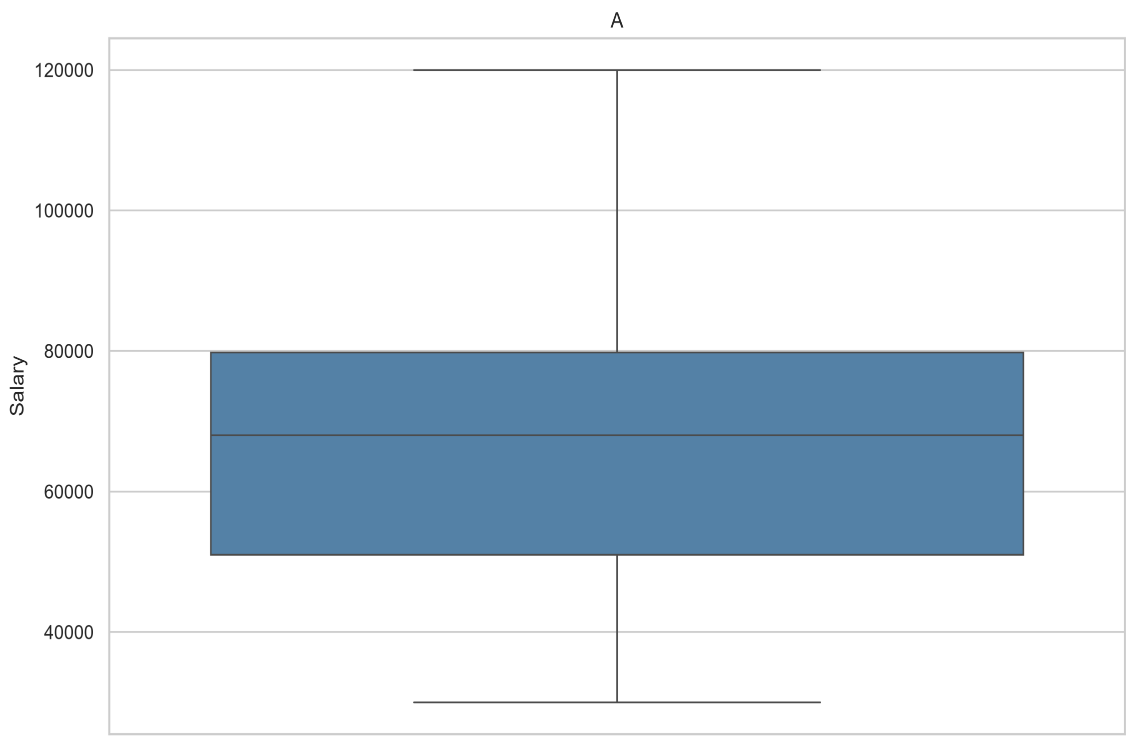
MISSING VALUES

Column	Missing
PassengerId	0
Name	0
Age	0
Gender	0
Department	0
Salary	1
Experience	0
JoiningDate	0
City	0
Performance	0

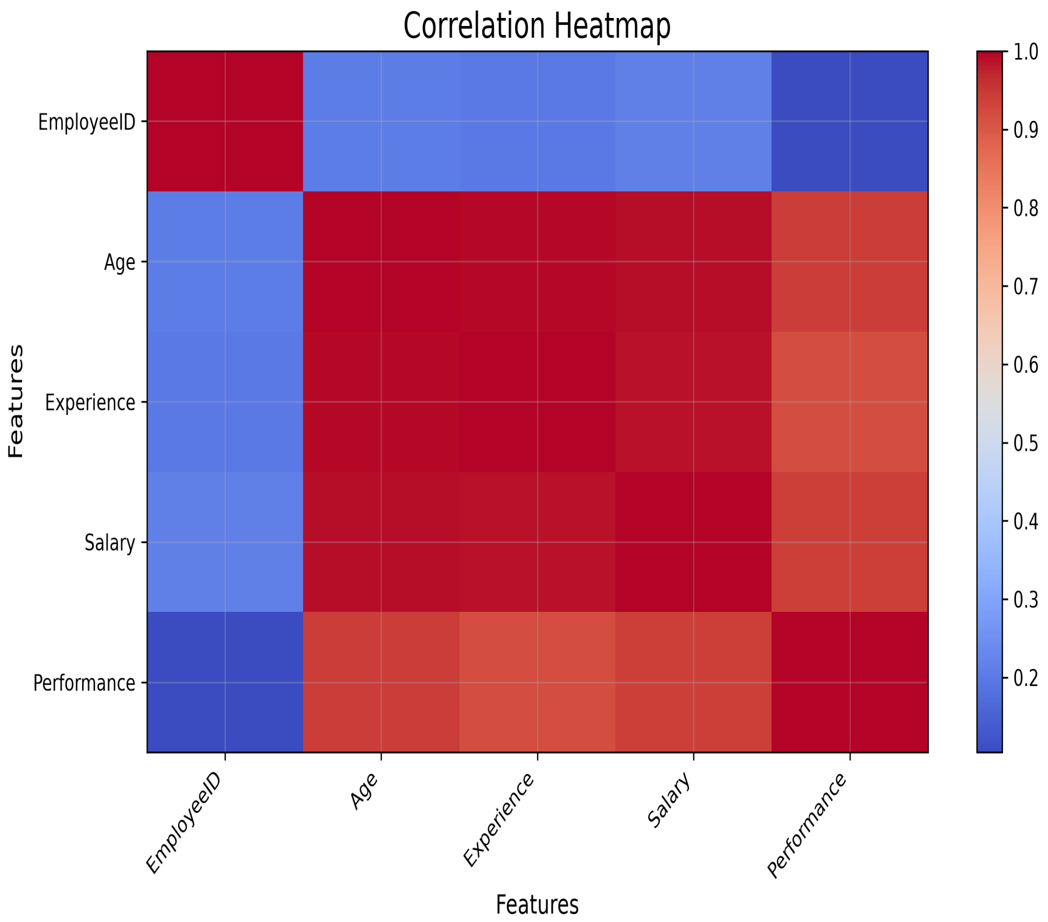
HISTOGRAM


Salary distribution
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Count
40000
60000
80000
100000
120000
Salary

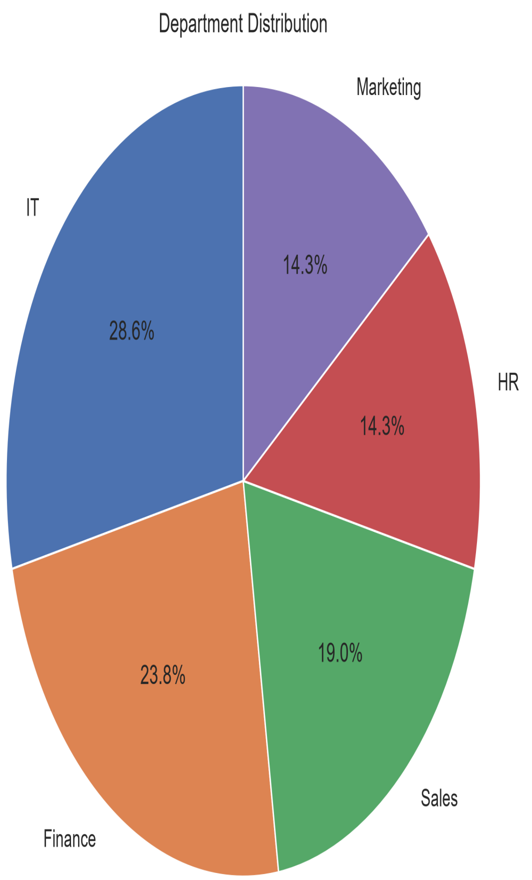
BOX PLOT


A
120000
100000
80000
60000
40000
Salary

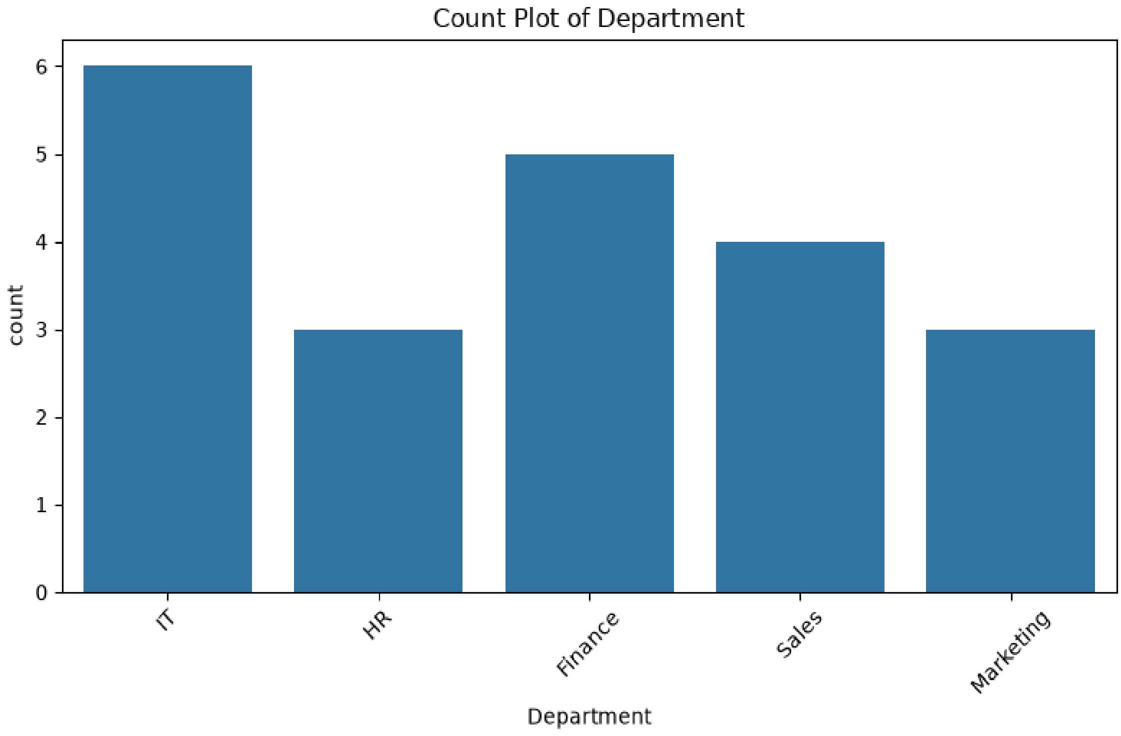
CORRELATION HEATMAP



PIE CHART


Department Distribution
Marketing
14.3%
IT
28.6%
HR
14.3%
19.0%
23.8%
Sales
Finance

COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
PassengerId	0
Age	0
Salary	2
Experience	1
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Salary	0.92
Age	Experience	0.96
Age	Performance	0.89
Salary	Experience	0.99
Salary	Performance	0.86
Experience	Performance	0.91

DATASET OVERVIEW

The dataset consists of 20 rows and 10 columns, capturing demographic data (Age, Gender, City), professional tenure (Experience, Joining Date, Department), and financial/performance metrics (Salary, Performance). An anomaly in the schema design is present, where the primary employee identifier is labeled PassengerId.

DATA QUALITY ANALYSIS

The overall dataset health score is high at 95. However, data completeness and consistency issues exist. Specifically, there is 1 missing value in the Salary column. Additionally, 3 potential outliers have been detected, comprising 2 anomalous records in Salary and 1 in Experience. No duplicate rows were identified.

STATISTICAL ANALYSIS

Statistical testing reveals extremely strong positive correlations across all numeric variables. Experience and Salary share an almost perfect correlation of 0.99, indicating a highly standardized, seniority-based pay scale. Age is highly correlated with Experience at 0.96 and Salary at 0.92. Notably, Performance strongly correlates with Experience at 0.91 and Salary at 0.86, showing that older, more experienced employees consistently receive the highest performance ratings.

BUSINESS INSIGHTS

The organization operates under a rigid, tenure-driven operational model. Compensation is strictly tied to years of experience rather than independent merit, which ensures predictable payroll scaling but may discourage fast-track promotions. Furthermore, performance scores increase directly with age and experience, suggesting that institutional knowledge is a key driver of employee productivity within departments like IT, HR, Finance, and Sales.

RISK ASSESSMENT

The 0.99 correlation between experience and salary poses a retention risk for high-performing junior staff, who may feel undervalued under a strictly seniority-based compensation structure.

Additionally, the missing salary value and outliers introduce risk to payroll budgeting and financial reporting accuracy if left unaddressed.

AI RECOMMENDATIONS

First, clean the dataset by imputing the single missing Salary value based on the employee's experience level and verifying the 3 outliers. Second, rename the PassengerId column to EmployeeId to align with professional data standards. Third, introduce performance-based incentives to supplement the rigid tenure-based compensation structure and retain high-potential junior talent.

Rule-Based Recommendations

✓ ■■ Dataset contains 1 missing values. Consider filling or removing them.

✓ ■ No duplicate rows found.

✓ ■■ 3 potential outliers detected.

✓ ■ Strong correlations detected. A Correlation Heatmap is recommended.

✓ ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.

✓ ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.

✓ ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.

CONCLUSION

The organization possesses a stable, highly experienced, and high-performing workforce with excellent overall data health. Resolving the minor missing data and outlier anomalies will secure the database integrity, while evolving the compensation model will help protect the talent pipeline from competitor poaching.